

Major Project End-Term Report

Project Title: UpTrend

A Web-Based Trading Education and Paper Trading Platform

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11th May, 2026

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CANDIDATE'S DECLARATION

I/We hereby certify that the project work entitled “**UpTrend**” in partial fulfilment of the requirements for the award of the Degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING with specialization in Graphics and Gaming and submitted to the Department of Systemics, School of Computer Science, University of Petroleum & Energy Studies, Dehradun, is an authentic record of my/ our work carried out during a period from **January, 2026** to **April,2026** under the supervision of **Dr Keshav Sinha**.

The matter presented in this project has not been submitted by me/ us for the award of any other degree of this or any other University.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Dr Keshav Sinha
Project Guide

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TABLE OF CONTENT

Sr. No.	Title
1.	Abstract
2.	Introduction
3.	Problem Statement & Objectives
4.	Literature Review
5.	System Architecture
6.	Methodology & Implementation
7.	Software Requirements
8.	Results & Final Output
9.	Testing & Validation
10.	Scope & Limitations
11.	Future Work
12.	Conclusion
13.	References

1.ABSTRACT

The rapid growth of online financial markets and digital trading platforms has encouraged a large number of beginners to participate in trading activities without proper education or practical exposure. Most novice traders rely on fragmented learning resources such as videos, blogs, and social media content, which often lack structure and practical application. As a result, beginners frequently experience confusion, emotional decision-making, and financial losses while entering real trading environments.

UpTrend is a web-based educational and paper trading platform developed to bridge the gap between theoretical financial learning and practical market experience. The platform combines structured learning modules, interactive quizzes, portfolio tracking, and a real-time paper trading simulator into a unified ecosystem. Users can learn trading concepts and immediately apply them in a risk-free environment using virtual money.

The application is developed using the MERN stack, including MongoDB, Express.js, React.js, and Node.js, along with Tailwind CSS for responsive user interface design. The platform includes modules such as user authentication, trading simulation, performance analytics, educational content, and portfolio management.

The primary goal of UpTrend is to provide beginners with a safe and interactive environment where they can understand market behavior, improve trading discipline, and build confidence before participating in real financial markets. The project demonstrates the effective integration of modern web technologies and financial simulation systems to create a scalable and user-friendly learning platform.

2. INTRODUCTION

Financial markets have become increasingly accessible due to the rapid advancement of digital technology and online brokerage systems. Today, individuals can participate in stock trading, cryptocurrency investment, and financial analysis directly through web and mobile platforms. Although this accessibility has created new opportunities for financial growth, it has also exposed beginners to significant challenges.

Most beginner traders enter financial markets without proper understanding of market behavior, risk management, or trading discipline. Educational resources available online are often unstructured and scattered across multiple platforms, making it difficult for learners to develop a systematic understanding of financial concepts. In many cases, users directly enter live trading environments without practical experience, leading to emotional decision-making and monetary losses.

Traditional methods of financial education focus heavily on theoretical explanations while lacking practical implementation. However, trading is a skill that requires experience, observation, analysis, and continuous practice. Without a safe environment for experimentation, beginners struggle to gain confidence and understand real-world market dynamics.

UpTrend was developed as a solution to this problem by integrating financial education with practical simulation. The platform provides a structured learning ecosystem where users can:

- Learn trading concepts through educational modules
- Complete quizzes to test understanding
- Practice trading using virtual money
- Analyze their performance through analytics dashboards

The project focuses on creating a beginner-friendly environment that promotes learning before execution. Instead of encouraging direct participation in live financial markets, UpTrend allows users to gradually build confidence through paper trading and guided educational content.

The platform also emphasizes user engagement through modern dashboard design, responsive interfaces, performance tracking, and gamified progress systems. By combining education, simulation, and analytics, UpTrend contributes toward improving financial literacy and responsible trading practices among beginners.

3. PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Beginner traders often face multiple difficulties while entering financial markets due to lack of practical exposure and structured educational guidance. Existing learning platforms mainly provide theoretical content without allowing users to apply concepts in realistic trading environments. Similarly, many trading platforms focus only on execution and ignore the educational needs of beginners. Some major challenges faced by beginner traders include:

- Lack of practical trading experience
- High financial risk in live markets
- Emotional trading decisions
- Absence of performance tracking systems
- Fragmented educational resources
- Difficulty understanding market behavior

These challenges create the need for a unified platform that combines education, simulation, and analytics into a single ecosystem.

Objectives

The major objectives of the UpTrend project are:

- To develop a paper trading simulator using virtual money
- To provide structured educational modules for beginners
- To implement portfolio management and profit/loss tracking
- To create interactive dashboards for analytics and monitoring
- To ensure secure user authentication and data management
- To improve financial literacy through practical learning
- To create a responsive and beginner-friendly user interface

The project aims to reduce the gap between theoretical learning and practical market understanding through simulation-based education.

4. LITERATURE REVIEW

The importance of financial literacy and simulation-based learning has significantly increased in recent years due to the rapid growth of online financial markets. Researchers and educational technology experts have emphasized the need for systems that combine theoretical understanding with practical application.

One of the most widely discussed approaches in financial education is Simulation-Based Learning (SBL). Studies suggest that simulations allow learners to practice complex decision-making skills in safe environments without facing real-world consequences. In trading education, paper trading platforms provide beginners with opportunities to understand market movement, risk management, and portfolio handling without financial loss.

Research on financial behavior also highlights the psychological challenges associated with trading. Emotions such as fear, greed, and overconfidence often influence beginner traders, resulting in impulsive decisions and inconsistent strategies. Several studies suggest that performance analytics and trading journals help users identify behavioral mistakes and improve discipline over time.

Interactive dashboards and data visualization tools have also become important in modern educational systems. Financial markets involve large amounts of dynamic data, which can become difficult for beginners to interpret. Visualization systems such as graphs, charts, and portfolio analytics improve understanding by simplifying complex numerical information.

Gamification is another important concept explored in educational technology research. Features such as quizzes, progress tracking, achievements, and levels increase user engagement and encourage consistent participation. In trading

education platforms, gamification can help users remain motivated while gradually improving their skills.

Modern web technologies such as React.js, Node.js, and MongoDB are increasingly used for scalable real-time applications due to their flexibility and performance capabilities. MERN stack applications are particularly suitable for interactive systems requiring continuous updates, secure authentication, and responsive interfaces.

Although several paper trading platforms exist, most focus only on simulation and lack integrated educational support. Similarly, educational platforms often fail to provide realistic practical environments for applying concepts. This gap creates the opportunity for systems like UpTrend, which combine learning, simulation, and analytics within a single platform.

The literature strongly supports the effectiveness of combining simulation systems, visualization tools, and structured learning environments to improve financial literacy and practical decision-making skills.

5. SYSTEM ARCHITECTURE

System Overview:

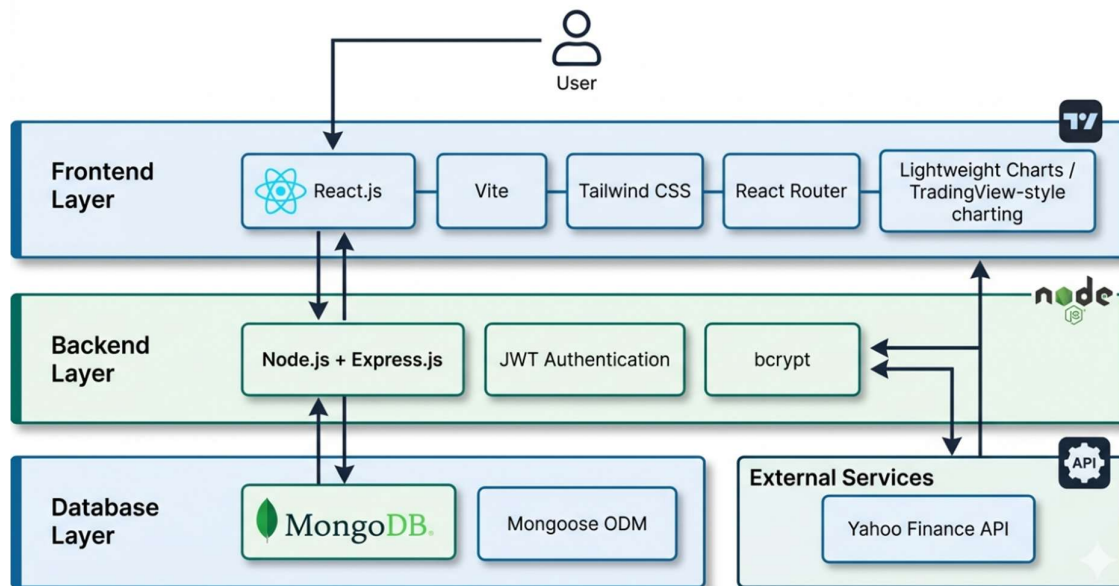


Fig 1: System Architecture of UpTrend

The UpTrend platform follows a three-tier architecture consisting of:

1. Frontend Layer

The frontend is developed using React.js and Tailwind CSS. This layer handles:

- User interface rendering
- Dashboard visualization
- Navigation between modules

- User interaction and responsiveness

2. Backend Layer

The backend is developed using Node.js and Express.js. It manages:

- Business logic
- Authentication systems
- Trading simulation logic
- API communication
- Portfolio calculations

3. Database Layer

MongoDB is used as the primary database for storing:

- User profiles
- Trade history
- Portfolio information
- Quiz progress
- Educational data

4. External Financial APIs

Financial APIs such as Yahoo Finance are integrated to fetch:

- Real-time stock prices
- Market updates
- Asset-related data

Core Modules

The major modules implemented in UpTrend include:

- User Authentication System
- Trading Simulator
- Portfolio Management
- Educational Academy
- Quiz & Assessment System
- Analytics Dashboard

6. METHODOLOGY & IMPLEMENTATION

The development of UpTrend followed an iterative and modular approach. The project was divided into frontend development, backend integration, database management, and testing phases.

Frontend Development

The frontend was built using React.js due to its component-based architecture and efficient rendering capabilities. Tailwind CSS was used for styling to create a modern and responsive user interface.

Features implemented in the frontend include:

- Responsive dashboards
- Trading interface

- Educational modules
- Charts and analytics
- Portfolio visualization

Backend Development

Node.js and Express.js were used to implement server-side logic. The backend handles:

- Authentication
- Trade execution
- Portfolio updates
- Profit/loss calculations
- Data retrieval

JWT-based authentication was implemented to ensure secure session management.

Database Management

MongoDB was used because of its flexibility and scalability. Collections were created for:

- Users
- Trades
- Holdings
- Learning progress
- Quiz results

Trading Simulation Logic

The trading engine allows users to:

- Buy assets
- Sell assets
- Track holdings
- Monitor virtual balance

Portfolio values are updated dynamically using market price data.

Educational Module

The academy section provides:

- Structured learning paths
- Topic-based lessons
- Interactive quizzes
- Progress tracking

Users are encouraged to complete educational modules before participating in simulated trading.

7. SOFTWARE REQUIREMENTS

Development Tools

- Visual Studio Code
- Git & GitHub
- Postman

Technologies Used

- React.js
- Node.js
- Express.js
- MongoDB
- Tailwind CSS

APIs & Libraries

- Yahoo Finance API
- Recharts
- JWT Authentication
- Axios

Hardware Requirements

- Minimum 8GB RAM
- Internet Connectivity
- Modern Web Browser

8. RESULTS & FINAL OUTPUT

The final implementation of UpTrend successfully demonstrates all planned functionalities and modules.

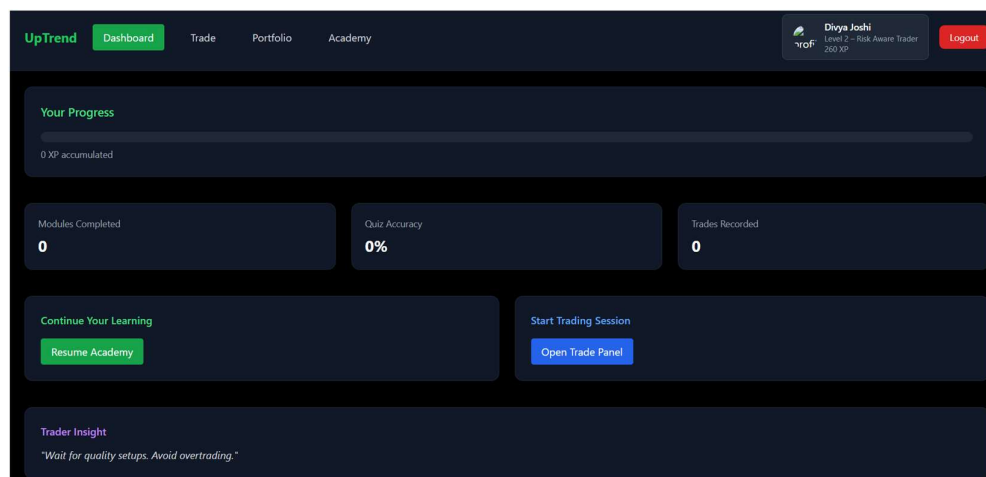


Fig 2: Dashboard

The dashboard displays:

- Trading capital
- Win rate
- Total trades
- Portfolio overview
- User activity analytics

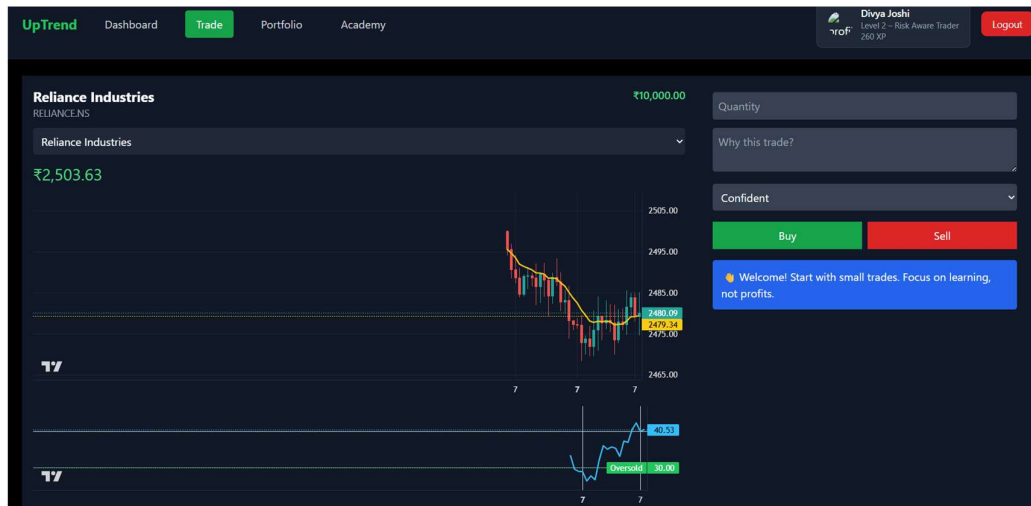


Fig 3: Trade Section

The trading section allows users to:

- Simulate buying/selling
- Monitor stock prices
- Execute paper trades
- Track virtual investments

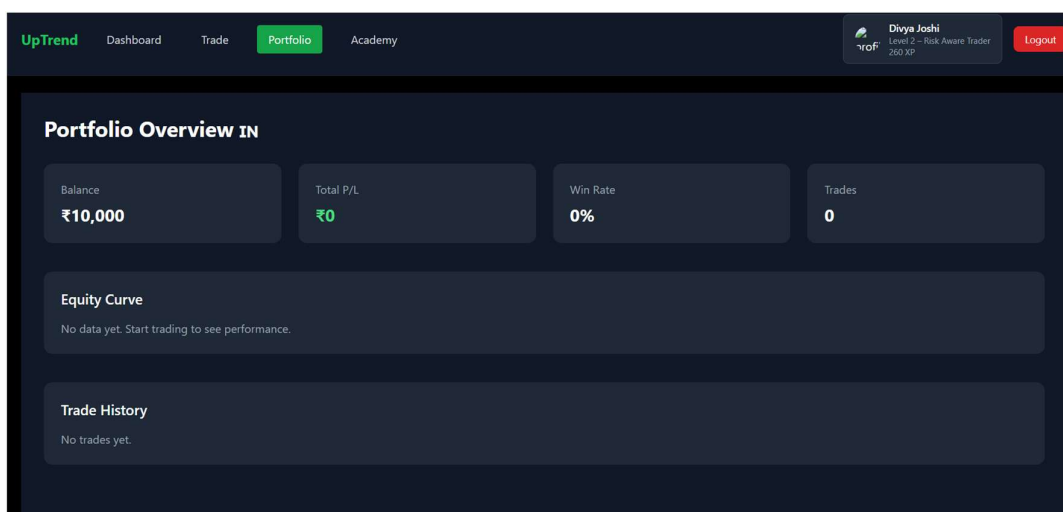


Fig 4: Portfolio Section

The portfolio module provides:

- Holdings overview

- Profit/Loss tracking
- Investment balance
- Asset allocation

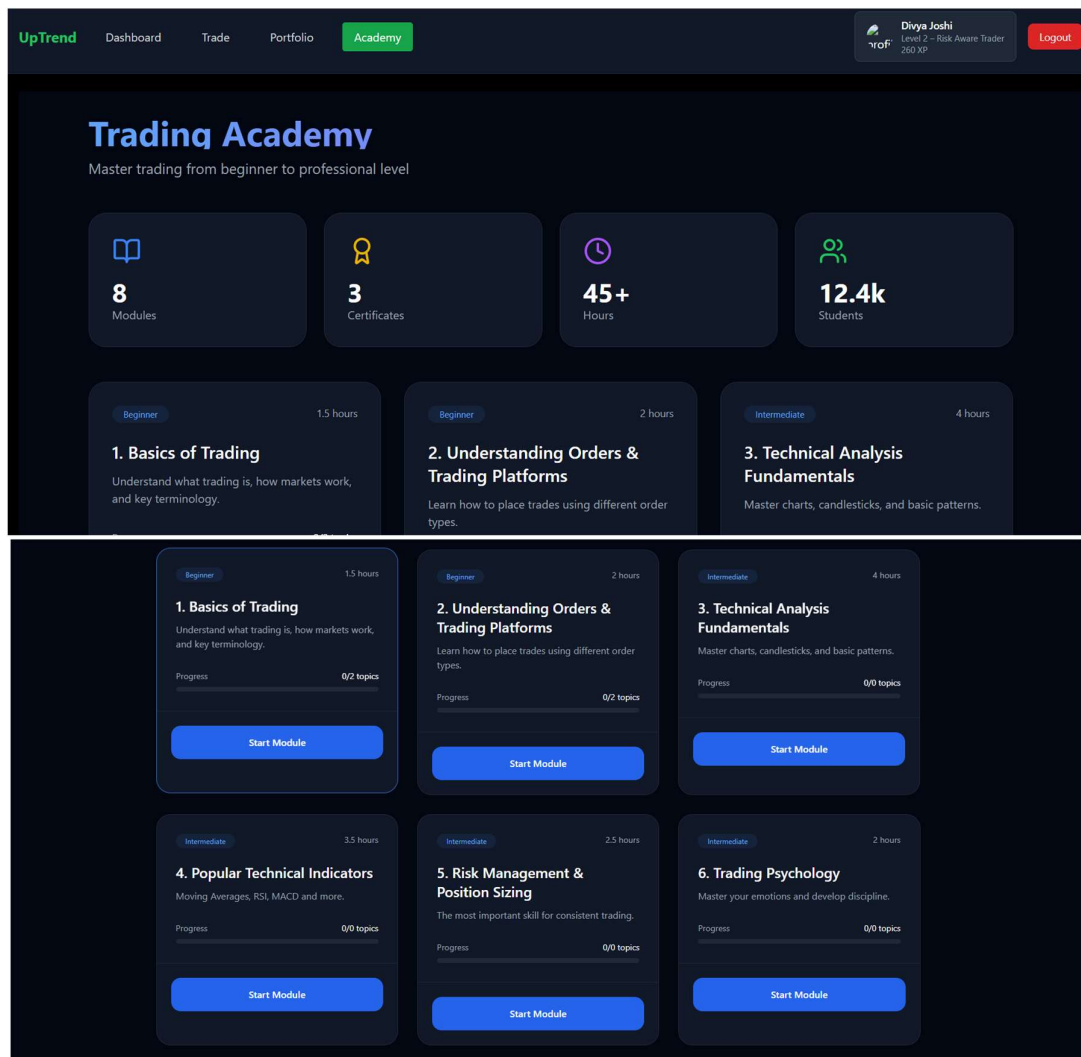


Fig 5: Academy Section

The academy module includes:

- Beginner trading concepts
- Market fundamentals
- Risk management lessons
- Interactive learning material

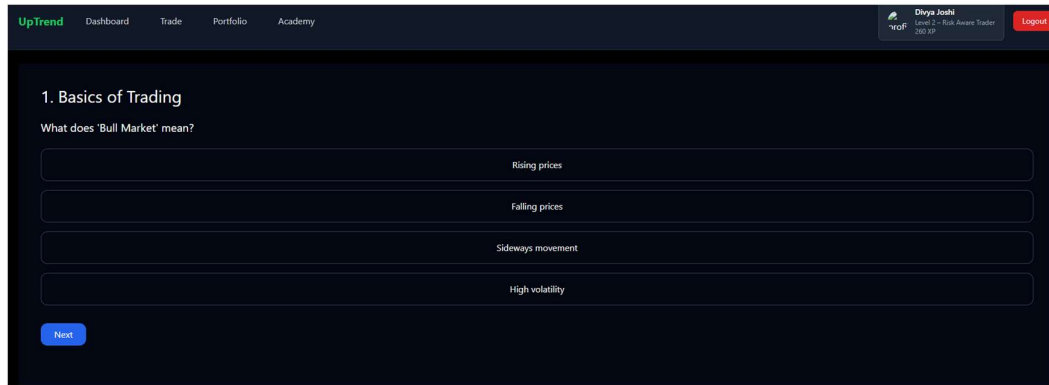


Fig 6: The quiz section validates user understanding before allowing progression to advanced modules.

9. TESTING & VALIDATION

Several testing methods were used during development to ensure system stability and functionality.

Testing Performed

Unit Testing

Verified correctness of:

- Profit/loss calculations
- Trade execution
- Portfolio updates

Integration Testing

Ensured smooth communication between:

- Frontend and backend
- APIs and database

UI Testing

Tested:

- Responsiveness
- Navigation
- User experience

Edge Case Testing

Handled:

- Invalid trades
- Empty portfolios
- Incorrect authentication attempts

The platform demonstrated stable performance and accurate functionality during testing.

10. SCOPE & LIMITATIONS

Scope

The scope of UpTrend includes:

- Financial education
- Trading simulation
- Portfolio analytics
- Beginner learning systems

The platform is designed mainly for educational purposes and beginner training.

Limitations

Some limitations of the project include:

- No live trading integration
- Possible delay in market data
- Limited advanced trading features
- Cannot fully replicate emotional pressure of real trading

11. FUTURE WORK

Several enhancements can be added in future versions of UpTrend:

- AI-powered trading assistant
- Real-time live trading integration
- Social trading community features
- Advanced technical indicators
- Leaderboards and competitions
- Mobile application support

These improvements can further enhance user engagement and learning experience.

12. CONCLUSION

UpTrend successfully demonstrates the integration of financial education and trading simulation into a unified web-based platform. The project addresses the challenges faced by beginner traders by providing a safe and interactive environment for learning and practice.

The platform combines structured educational content, portfolio management, analytics, and paper trading to create a comprehensive beginner-friendly ecosystem. Through simulation-based learning, users can improve their understanding of market behavior, trading discipline, and decision-making skills without financial risk.

The project also showcases the effective use of modern web technologies such as React.js, Node.js, Express.js, and MongoDB in developing scalable and responsive applications. Overall, UpTrend represents a practical and impactful

approach toward improving financial literacy and responsible participation in financial markets.

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